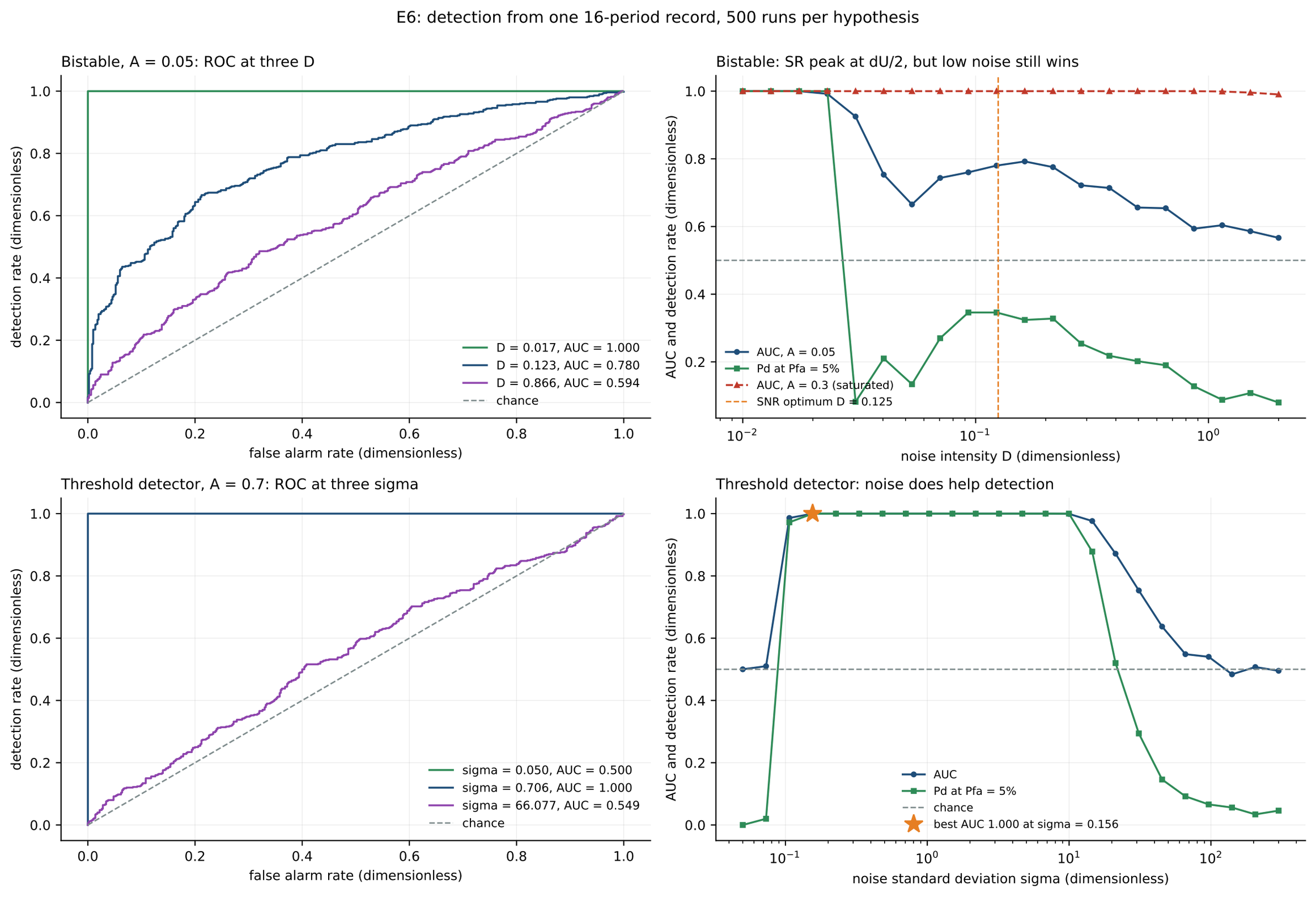
E6: detection from one 16-period record, 500 runs per hypothesis
Bistable, A = 0.05: ROC at three D
detection rate (dimensionless)
false alarm rate (dimensionless)
D = 0.017, AUC = 1.000
D = 0.123, AUC = 0.780
D = 0.866, AUC = 0.594
chance
Bistable: SR peak at dU/2, but low noise still wins
AUC and detection rate (dimensionless)
noise intensity D (dimensionless)
AUC, A = 0.05
Pd at Pfa = 5%
AUC, A = 0.3 (saturated)
SNR optimum D = 0.125
Threshold detector, A = 0.7: ROC at three sigma
detection rate (dimensionless)
false alarm rate (dimensionless)
sigma = 0.050, AUC = 0.500
sigma = 0.706, AUC = 1.000
sigma = 66.077, AUC = 0.549
chance
Threshold detector: noise does help detection
AUC and detection rate (dimensionless)
noise standard deviation sigma (dimensionless)
AUC
Pd at Pfa = 5%
chance
best AUC 1.000 at sigma = 0.156